

MTH2021 Mathematical Methods 2

Homework set 2

1. Differentiate $f(z) = z^3$ from first principles (i.e. starting from the definition of derivative).
2. Show that each function below is non-analytic by identifying two independent directions, along which the rates of change of the function are different:

$$(a) \quad \operatorname{Re} z \quad (b) \quad |z|^2 \quad (c) \quad \frac{z}{z^*}$$

3. Use the Cauchy-Riemann equations to determine whether, and where, the following functions are analytic:

$$\begin{array}{lll} (a) \quad z^3 & (b) \quad 1/z & (c) \quad \operatorname{Re} z \\ (d) \quad \exp z & (e) \quad z^* & (f) \quad \sin z \\ (g) \quad \cosh z & & \end{array}$$

4. Using the definitions of the circular and hyperbolic functions, show that

$$\begin{array}{ll} (a) \quad \sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2 \\ (b) \quad \cosh(z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2 \\ (c) \quad \sin(iz) = i \sinh z \\ (d) \quad \cosh(iz) = \cos z \end{array}$$

Evaluate $\sin(i\pi/2)$, $\sinh(i\pi/2)$, $\cos(i\pi)$, $\cosh(i\pi)$.

5. Calculate the principal value of $\log z$ (denoted by $\operatorname{Log} z$) for z given by

$$(a) \quad 1 + i \quad (b) \quad 2 - 2i \quad (c) \quad -5$$

6. Find all distinct solutions for z of the equations

$$(a) \quad z = \log(-e^2) \quad (b) \quad \exp z = -1 + i \quad (c) \quad \exp(iz) = \sin(2i)$$
